



Operator's Manual for Attachment



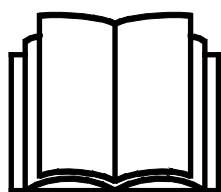
Silage fork

Product number

800 mm	A428328
900 mm	A36299
1050 mm	A36313
1300 mm	A36319

Including options

Silage dozer blade	A37111
Feeding table cleaning blade	A2936
Push broom adapter	A447080
Mounting plate	A36337



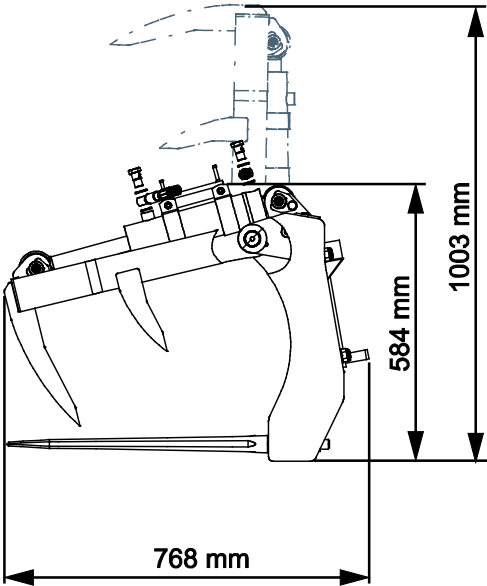
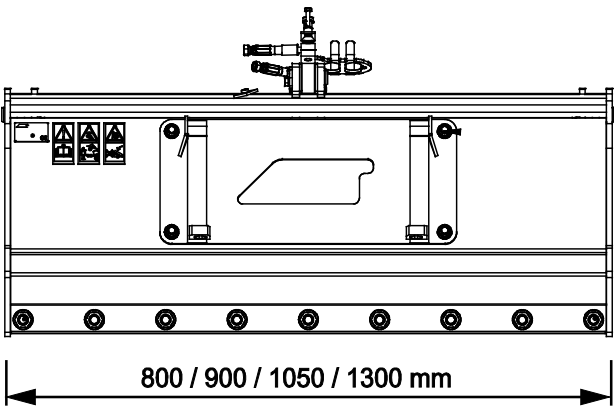
Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

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I. Foreword

Avant Tecno Oy would like to thank you for your purchase of this attachment for your Avant loader. It has been designed and manufactured based on years of experience on product development and manufacturing. By familiarizing yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

The purpose of this manual is to help you to:

- operate the equipment in a safe and efficient manner
- observe and prevent any hazardous situations
- keep the equipment intact and ensure a long service life

With these instructions, even an inexperienced user can use the attachment and loader safely. The manual includes important instructions for experienced Avant operators as well. Ensure that all persons using the loader have received proper guidance and familiarised themselves with the manual of the loader, each attachment that are used, and all safety instructions before using the equipment. Using the equipment for other purposes or use in any other way than described in this manual is prohibited. Keep this manual at hand throughout the service life of the equipment. If you sell or transfer the equipment, be sure to hand over this manual to the new owner. If the manual is lost or damaged, you can request a new one from your Avant dealer or from the manufacturer.

In addition to the safety instructions included in this Operator's Manual, you must observe all occupational safety regulations, local laws, and other regulations concerning the use of the equipment. Particularly the regulations concerning the use of the equipment on public road areas must be observed. Contact your Avant dealer for more information about local requirements before you operate the loader on road areas.

Contact your local Avant dealer for any questions, service, spare parts or about any problems that may occur with the operation of your machine.

Due to continuous product development some of the details shown in this manual may differ from your equipment. Pictures may also depict optional equipment or features that are not currently available. We reserve the right to change the contents of the manual without notification. Copyright © 2024 Avant Tecno Oy. All rights reserved.

Warning symbols used in this manual

The following warning symbols are used throughout this manual. They indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:



WARNING SAFETY ALERT SYMBOL

This symbol means: **“Warning, be alert! Your safety is involved!”**

Carefully read the message that follows, it warns of an immediate hazard that could cause serious personal injury.

The safety alert symbol by itself and with related safety statement indicates important safety messages throughout this Manual. It is used to draw attention to instructions involving your personal safety or the safety of others. When you see this symbol, be alert, your safety is involved, carefully read the message that follows, and inform other operators.

DANGER

This signal word indicates a hazardous situation which, if not avoided, will cause death or serious injury.

WARNING

This signal word indicates a potentially hazardous situation which, if not avoided, could cause serious injury or death.

CAUTION

This signal word is used when minor injury could result if the instructions are not followed properly.

NOTICE

This signal word indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

2. Designed purpose of use

The Avant silage fork is an attachment that is designed and manufactured to be used with Avant compact loaders that are shown in Table 1. The silage fork is intended to move or distribute silage or similar loose materials, where the grapple of the attachment ensures a good grip of the material. The silage fork is also suitable for transporting bales, depending on the load carrying capacity of the loader.

The lower tines of the silage fork are made of reinforced steel and they are replaceable. In standard configuration, the silage fork has been designed to avoid contact with the front tyres of the loader. In applications where increased tipping angle is necessary, the mounting plate A36337 can be used, except with the smallest forks model. When the adapter is used, it is the responsibility of the operator to avoid contact with the front tyres of the loader.

The silage fork can be equipped with an optional silage dozer blade A37111. The silage dozer blade is mounted on the silage forks, and it is intended for pushing the uneaten silage back to the cattle on the feeding table.

A feeding table cleaning blade A2936 with a rubber edge is also available as an option. The cleaning blade makes it easy to clear feeding tables before distributing fresh silage. The cleaning blade fits all forks and it can be attached without leaving the driver's seat.

The silage fork accessories are not designed for any other use than what is specified in this manual and must not be used for any other purposes than what they are intended for.

The attachment has been designed to require as little maintenance as possible. The operator can perform regular maintenance tasks. All repair work can't be performed by the operator, and demanding repair and maintenance operations are to be left for professional technicians. All maintenance work must be done using proper safety equipment. Spare parts must be identical with original specifications, which can be ensured by using only original spare parts. A separate spare parts catalogue may be available, consult your Avant dealer.

Familiarise yourself with the manual's instructions regarding service and maintenance. Please contact your Avant dealer if you have additional questions about the operation or maintenance of the equipment, or if you require spare parts or maintenance services.

Table 1 - Silage fork - Compatibility with Avant loaders

Avant		220 series 2 225 225LPG	313S 320S	419 420 423 520 523	525LPG 528 530	630 635 635i 640 640i 645i 650i	735 735i 745 750 755i 760i	845 850 855i 860i	e5 e6 e513 e527
A428328	800 mm	(●)	•	•	•	•	(●)	(●)	•
A36299	900 mm	(●)	•	•	•	•	(●)	(●)	•
A36313	1050 mm	(●)	•	•	•	•	(●)	(●)	•
A36319	1300 mm	(●)	•	•	•	•	(●)	(●)	•

The compatibility recommendations are based on the width of the loader, lifting capacity, and adequate breakout force. Normally the width of the attachment should match roughly the width of the loader. When using too wide attachment the actual breakout force (bucket tilting capacity) decreases and the overall performance will decrease.

It is recommended to use the attachment only with loaders marked with • in Table 1. When using with models marked with (●), full performance may not be achieved. Additional counterweights may also be required. For compatibility with a model not shown in table, contact your Avant dealer.

NOTICE

Even though you can use the silage forks with 700 series loader models, it is recommended to use XL silage grab with these loaders. Other silage fork models can get damaged when you use them on 700 or 800 series loaders.

3. Safety instructions for using the attachment

Safety is the result of several factors. The combination of a loader and an attachment is highly powerful, and improper or careless use or maintenance may cause serious personal injury or property damage. Because of this, each operator must read and understand the operator's manuals for both the loader and the attachment before operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards.



DANGER

Misuse, careless use, or using an attachment that is in poor condition may cause risk of serious injuries. Familiarise yourself with the controls of the loader, correct coupling procedure, and the correct way to operate the attachment at a safe area. Study especially how to stop the equipment in a safe manner. Read all safety precautions carefully.

Read all safety instructions carefully before handling the attachment



WARNING

- When attaching an attachment to the loader, **make sure that the locking pins of the loader's quick coupling plate are in the lower position and that they have locked the attachment to the loader.** Never lift or move an unlocked attachment.
- This attachment is designed to be used by one operator at a time. Do not let others near the danger area of the equipment when it is in use.
- Always transport the attachment as low as possible to keep the centre of gravity low, and keep the telescopic boom retracted during driving.
- **Risk of crushing - The sharp tines of the attachment and its grapple can cause serious injuries.** Watch out for the sharp tines and the grapple, which may lower even if the loader is not running. Lower the attachment on the ground and lower its grapple completely before leaving the driver's seat to avoid unintentional movements of the loader boom, the attachment, and the grapple. Follow safe stopping procedure.
- **Sudden movements can cause the loader to tip over - Operate the controls of the loader in a slow and calm manner.** Be careful when lifting load to high level or lifting load from high. Avoid sudden changes in speed or direction to maintain balance of the loader especially when handling heavy loads. Drive slowly and carefully especially on inclined terrain or slippery surfaces.
- **Risk of tipping over - Keep loads close to ground while driving.** Carrying heavy loads can shift the centre of gravity of the loader and lead to tipping over of the loader. For the best stability always transport loads as low and close to the loader as possible, with the telescopic boom completely retracted, to keep the centre of gravity low.
- **Risk of crushing - Never go or allow anyone else to go under the load, the loader boom, or the attachment.** The stability of the loader may change when leaving the driver's seat, leading to the machine tipping over. Always remember that the boom may lower unexpectedly because of loss of stability, mechanical fault, or if another person operates the controls of the loader, leading to crushing hazard. The loader is not intended to be left to keep a load elevated for longer periods. Lower the attachment firmly on the ground before leaving the driver's seat and never allow anyone to be under the boom or the attachment.
- **Risk of falling objects - Never exceed lift capacity of the loader.** Observe maximum load indicated in the operator's manual of the loader. Be especially careful when the load sensor indicator is activated.



- **Risk of collision - Keep a safe distance to other people and obstacles in the operating environment.** Pay attention to the surroundings and any other persons and machines moving in the vicinity. Other persons must keep safety distance of 2 meters. Pay attention to the contours of the terrain and other hazards, such as branches and trees that can reach to the driver's area, loose rocks, and slippery surfaces.
- **Risk of suffocation - Never operate the loader indoors or at poorly ventilated areas.** Ensure that ventilation is sufficient when operating indoors. Do not operate the loader in closed spaces regardless of the engine or fuel type. Exhaust gases may concentrate to hazardous levels.
- Keep the loader articulation in straight position when handling heavy loads. When turning the articulation, the loader may tilt forward and tip over.
- Make sure that the surface can bear the total load. Also follow the correct tyre pressure settings.
- Observe maximum load indicated in the operator's manual of the loader. Be especially careful when the load sensor indicator is activated, if your loader is fitted with a load sensor system.
- Make sure that overhead clearance is sufficient. Hitting an overhead obstacle may cause the loader to tip over. Keep a safe distance from electric cables, lamps, or other electric systems; hitting live parts may cause electric shock.
- Never use the attachment to lift or to transport persons or as any kind of work platform even temporarily.
- Make sure to use only an attachment that is in good condition. Check the attachment thoroughly in regular intervals. Do not modify the attachment in a manner that would affect its safety. It is prohibited to drill holes on the attachment, and welding or other means of attaching hooks or other objects on the attachment is strictly prohibited.
- Shut down the loader and place the attachment to a safe position as shown in Safe stopping procedure before any cleaning, maintenance, or adjustments.
- Use the equipment only for its intended purpose. Other use may create unnecessary safety risks, and the equipment may get damaged.
- Make sure that the loader is equipped with necessary safety components, and that they are in working condition. Seat belt must be used. If there are specific hazards related to the operating area, use appropriate safety equipment.
- Also read the safety instructions and correct use of the loader from the operator's manual of the loader.

**DANGER**

Risk of crushing - Going under the loader boom or an attachment can cause serious injury or death. Never go under the loader boom and prevent others from getting near lifted boom or attachment. Going under a raised attachment or loader boom is dangerous, as the boom may lower due to loss of stability, mechanical fault or when other person operates the controls of the loader. Lower the attachment firmly on the ground to a safe position before leaving the driver's seat.

Keep in mind that the loader boom can be lowered or tilted even if the engine has been shut down (crushing hazard). The loader is not intended to be left to keep a load elevated for longer periods. Always lower the attachment to a safe position before leaving the driver's seat.

**WARNING**

Crushing hazard - Never allow anyone to get under a lifted attachment or loader boom. Keep in mind that the loader boom can be lowered or tilted even if the engine has been shut down. The loader is not intended to be left to keep a load elevated for longer periods. Always lower the attachment to a safe position before leaving the driver's seat.



3.1 Safe shutdown procedure

Safe stopping of the attachment, before going near the attachment:

**WARNING**

Always stop the attachment following safe stopping procedure before leaving the driver's seat. Safe stopping procedure prevents all unintentional movements of the attachment. Note that the loader boom can move down even if the engine of the loader is turned off. Safe stopping procedure:

- Lower the boom and the attachment on the ground.
- Shut down the loader engine and lock the parking brake.
- Release residual pressure from the hydraulic system; move all hydraulic control levers to their extreme positions a couple of times.
- Prevent starting of the machine, remove ignition key.
- Make sure the grapple is completely lowered before leaving the attachment unattended.

3.2 Personal protective equipment

Remember to wear proper personal protective equipment:



- The noise level at the driver's seat may exceed 85 dB(A) depending on loader model and operating cycle. Extended exposure to loud noise can cause hearing impairment. Wear hearing protection while working with the loader.



- Wear protective gloves.



- Wear safety boots whenever working with the loader.



- Wear safety glasses e.g. when handling hydraulic components.

4. Technical specifications

Table 2 – Silage forks - Specifications

<i>Product number</i>	<i>A428328</i>	<i>A36299</i>	<i>A36313</i>	<i>A36319</i>
Working width:	800 mm	900 mm	1050 mm	1300 mm
Weight:	98 kg	126 kg	138 kg	160 kg
Other dimensions:	See Overview			
Number of tines, grapple/frame:	5/6	5/6	6/7	7/9
Maximum input of hydraulic energy:	50 l/min, 22,5 MPa (225 bar)	30 l/min, 20 MPa (200 bar)		
Compatible Avant loaders:	See Table 1, Designed purpose of use			
<i>Available options</i>				
Mounting plate*:	A36337 Increases tilting angle by 17°			
Push broom adapter:	A447080			
Feeding table cleaning blade:	A2936 Width 1200 mm Side extension width 550 mm Weight 50 kg			
Silage dozer blade:	A37111 Blade length 1800 mm Working width 1160 mm Weight 92 kg			

*

NOTICE

The mounting plate is not suited for use with all loader models. It is intended for use in special cases, where more forward tilting is necessary. When using the mounting plate, the front tyres may be damaged, as the tines will reach the tyres. Check the compatibility with your AVANT dealer.

4.1 Safety labels and main components of the attachment

Listed below are the labels and markings on the attachment. They must be visible and readable on the equipment. Replace any unclear or missing label. New labels are available via your retailer or contact information provided on the cover.



WARNING

Make sure all warning decals are readable. The warning labels contain important safety information and they help to identify and remember the hazards related to the equipment. Replace damaged or missing warning labels with new ones.



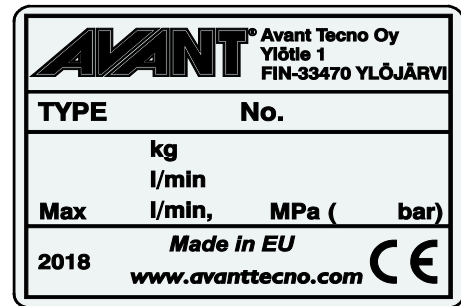
A46771



A46772



A46803



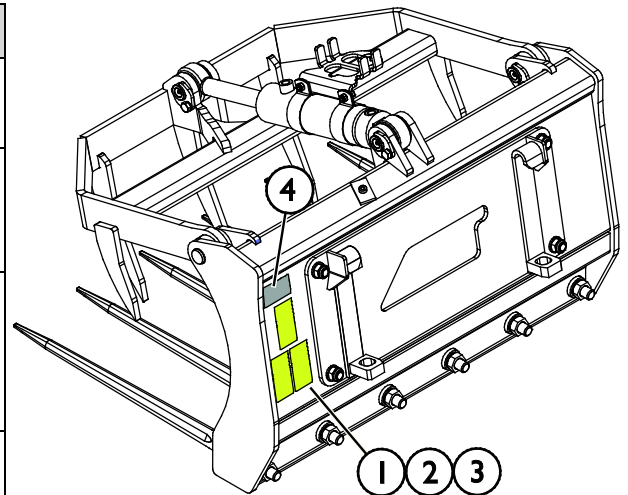
Attachment identification plate

A428335, A417966, A417973, A418022

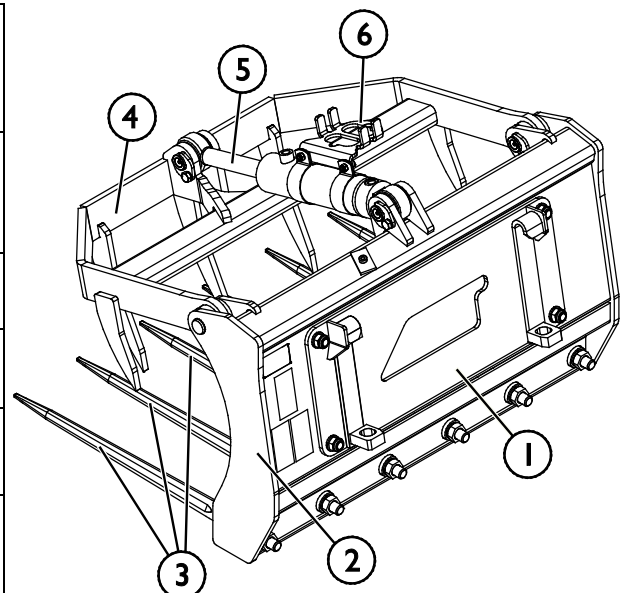
To apply a new label: Clean the surface thoroughly to remove all grease. Allow it to dry completely. Peel off the protective film of the label and press firmly to apply. Take care not to touch the glue of the label.

Table 3 - Decal locations and warning messages

	Decal	Warning message
1	A46771	Misuse hazard - Read instructions before use.
2	A46772	Crushing hazard - Do not go under a raised attachment; stay away from raised equipment.
3	A46803	Pinching and cutting hazard, keep clear of moving parts, do not leave the equipment running. Operate the attachment only from the driver's seat.
4	A423101	Attachment identification plate


Table 4 - Silage fork - Main components

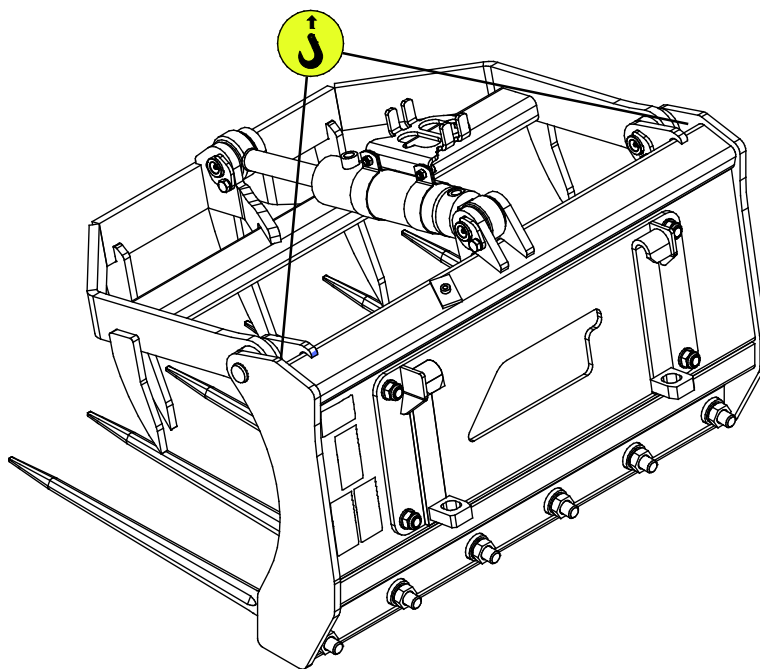
1	Quick coupling plate: Standard: A417911 Mounting plate with increased tipping angle: A36337
2	Frame
3	Replaceable tines
4	Grapple
5	Grapple cylinder
6	Multi connector holder



4.2 Lifting

Make sure that the attachment is properly secured against unintended movements during lifting operation. Use appropriate type of lifting accessories, lifting straps, chains or hooks, for example. Do not use lifting accessories if there are signs of damage, deformation, or otherwise unsafe conditions.

Lift the attachment by using the lifting points, that are shown in the adjacent figure.



5. Attaching the attachment

Attaching the attachment to the loader is quick and easy, but it must be done carefully. The attachment is mounted to the loader boom by using the quick coupling plate on the loader boom and the counterpart on the attachment.

If the attachment is not locked to the loader, it may fall from the loader and cause a hazardous situation. Never drive with the loader and never lift the loader boom if the attachment has not been fully locked. To prevent hazardous situations, always follow the coupling procedure shown below. Also remember the safety instructions shown in this manual.



Crushing risk - Make sure that an unlocked attachment will not move or fall over.

Do not stay in the area between the attachment and the loader. Mount the attachment only on level surface.

Never move or lift an attachment that has not been locked.

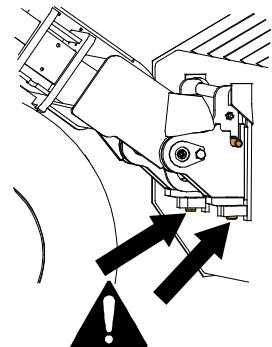


Risk of crushing - Avoid tipping over the attachment. Excessive tilting or lifting of an unlocked attachment increases the risk of tipping the attachment over. Do not use the automatic locking of the locking pins when the attachment is lifted more than one meter from the ground. If the locking pins do not return to the normal position when tilting, do not tilt or raise the attachment any more. Lower the attachment on the ground and secure the locking manually.



Risk of falling objects - Prevent dropping of attachment.

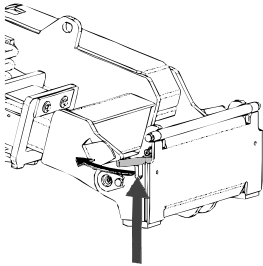
An attachment that has not been completely locked to the loader may fall on the boom or towards the operator, or fall under the loader during driving, causing loss of control of the loader. Never move or lift an attachment that has not been locked. Before moving or lifting the attachment, make sure that the locking pins are in the lower position and come through the fasteners on the attachment on both sides.



Make sure the loader is compatible with the attachment. Stability of the loader, possible overload of the attachment, and compatibility of loader control systems must be ensured in addition to mechanical compatibility of the attachment. If you use the attachment with a loader that is not fit to be used with the attachment model you have, risks include tipping over, damage to attachment due to overload, and risks related to uncontrolled motion of attachment and its parts. If your loader is not listed in Table 1 on page 6, ask your Avant dealer before using this attachment.

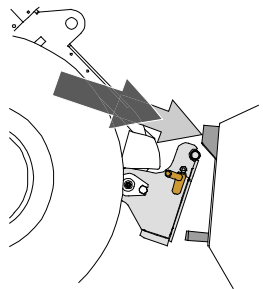
Avant quick coupling system:

Step 1:



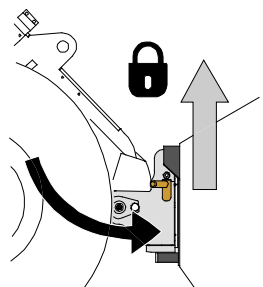
- Lift the locking pins of the quick coupling plate of the loader up, and turn them backwards into the slot, so that they are locked in the upper position.
- If your loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the operator's manual of the loader.
- Make sure that the hydraulic hoses are positioned so that they will not get in between the coupling plate of the loader and the attachment, and that they will not get squeezed or chafed against sharp edges.

Step 2:



- Turn the quick coupling plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.
- Align the upper pins of the loader's quick coupling plate so that they are under the corresponding brackets of the attachment.

Step 3:



- Lift the boom slightly – pull the boom control lever backward to lift the attachment just off the ground.
- Turn the boom control lever left to turn the bottom section of the quick coupling plate of the loader onto the attachment.
- Lock the locking pins manually or lock the hydraulic locking.
- **Always check the locking of both locking pins.**

5.1 Connecting and disconnecting hydraulic hoses

On Avant loaders the hydraulic hoses are connected using the multi connector system. If you have an older model Avant 300-700 series loader with the conventional quick couplers, and you want to change to the multi connector system, contact your Avant dealer or service point for instructions or installation services.



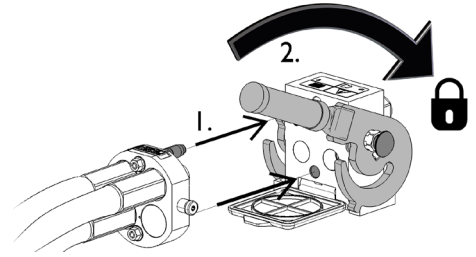
Risk of movement of the attachment and ejection of hydraulic oil - Never connect or disconnect quick couplings or other hydraulic components while the auxiliary hydraulics control lever is locked on or if the system is pressurized. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment, or ejection of high-pressure fluid, which can cause serious injuries or burns. Follow safe stopping procedure before disconnecting hydraulics.

NOTICE

Keep all fittings as clean as possible; use the protective caps on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place the couplings onto the holder on the attachment.

Connecting the multi connector system:

1. Align the pins of the attachment connector with corresponding holes of the loader connector. The multi connector will not connect if the attachment connector is upside down.
2. Connect and lock the multi connector by turning the lever towards the loader.

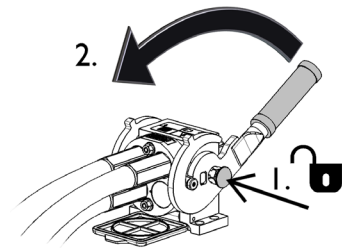


The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Also shut down the loader and release the residual hydraulic pressure.

To disconnect the multi connector system:

Before disconnecting put the attachment down on a solid and even surface.

1. Switch off the auxiliary hydraulics of the loader.
2. While pushing unlock button, turn the lever to disconnect the connector.
3. After ending operation put the multi connector on its holder on the attachment.

**Disconnecting hydraulic hoses:**

Before disconnecting the fittings, lower the attachment to safe position on solid and level surface. Turn the control lever of the auxiliary hydraulics to its neutral position.

NOTICE

When uncoupling the attachment, always disconnect the hydraulic couplings before unlocking the quick coupling plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.

Releasing residual hydraulic pressure:

In case residual pressure is left in the hydraulic system of the attachment, it is often possible to disconnect the hydraulic couplings, but it may be difficult to connect them the next time. If the fittings will not connect, the residual pressure must be released by turning the auxiliary hydraulics control lever of the loader, when the engine is turned off. To make sure that there will not be residual pressure in the hydraulic system of the attachment, shut down the loader engine and move the auxiliary hydraulics control lever of the loader back and forth before disconnecting the couplings.

6. Instructions for use

Practise the use of the attachment and the controls of the loader in a safe area. If you are unfamiliar with the particular loader model, it is recommended to practice its use without any attachments.



Risk of tipping over - Avoid overload. Loader can tip over when handling too heavy loads, or because of dynamic movements caused by driving with and handling of a heavy load. Do not extend the telescopic boom when the load approaches the lifting capacity of the loader, or when the loader boom is in horizontal position. When you notice that the rear wheels of the loader are about to lift from the ground, or if the load indicator of the loader (if fitted) signals:

- Lower the load calmly and retract the telescopic boom
- Avoid sudden changes in speed or direction whenever load is lifted
- Use additional counterweights as needed



Risk of falling from height and crushing under an attachment or loader - Never use the attachment as a lifting platform or to carry people. Never use the loader or its attachments to lift persons or as any kind of work platform even temporarily. Never climb on the attachment.

Always lower the attachment on the ground before leaving the driver's seat. The loader is not intended to hold loads lifted for longer periods of time.



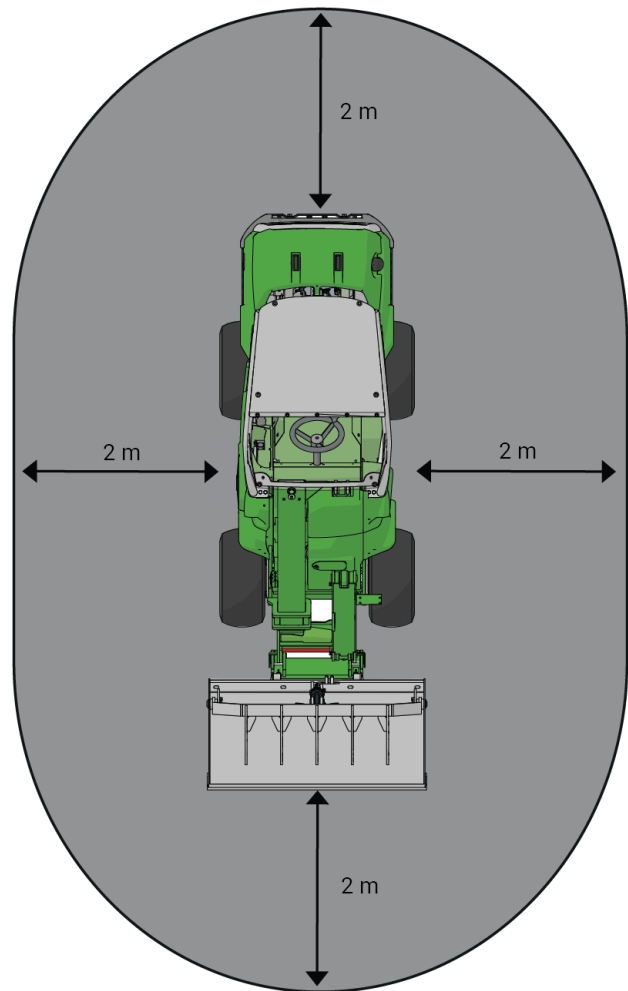
Risk of crushing - The sharp tines of the attachment and its grapple can cause serious injuries. Make sure that there are no other persons at the operating area and keep a safe distance to persons or farm animals. Lower the grapple before driving with the loader. Stop the loader following the safe stopping procedure before going near the attachment.



6.1 Safety distance

The minimum safety distance from the attachment is 2 metres. Stop the attachment immediately if bystanders get within the safety distance.

Do not let bystanders get any closer than 2 metres from the attachment. Also have another person watching that bystanders do not get too close.



6.2 Checks before use

- Check the operation of the grapple before each work shift:
 - Check that the hydraulic hoses do not chafe.
 - Check that the grapple joints are lubricated.
- Check that the coupling plate is correctly attached to the silage forks.
- Make sure that bystanders are at a safe distance when operating the equipment. Do not let anyone enter danger area of the boom or to stay directly in front of the loader. Also make sure that it is safe to reverse with the loader. Do not assume that bystanders will remain where you last saw them. Especially children are often attracted to moving equipment.
- Check the general condition of the attachment and the loader, and check for possible hydraulic oil leaks. Do the daily checks of the loader, see the operator's manual of the loader. The attachment must not be used if there is a fault in the hydraulic system of the loader or the attachment. Refer to Chapter 7 for maintenance instructions.
- Operate the attachment and the controls of the loader only when seated on the driver's seat. Ensure that the loader and the attachment are used in a safe manner and as intended. Do not allow children to operate the equipment.
- Remember correct and healthy working methods and avoid leaving the driver's seat unnecessarily.

6.3 Operator qualification

Anyone who intends to use this attachment must:

- Know the intended use of this attachment.
- Know how to use the loader safely in different operating conditions.
- Read and always follow the instructions concerning the use of the attachment in this operator's manual.
- Have this manual and the operator's manual of the loader available.
- Ensure that the loader and the attachment are used in a safe manner and as intended.
- Do not allow children to operate the equipment.
- Do not operate the loader or attachments while under the influence of alcohol, drugs, medication that may impair judgement or cause drowsiness, or if not otherwise medically fit to operate the equipment.
- Complete any possible mandatory training required by the employer before use.

6.4 Operating the silage fork



WARNING

Risk of serious personal injuries - Ensure that there are no other persons near the operating area. Do not allow others closer than 2 metres from the attachment, to the boom danger area, or directly in front or back of the loader. Move carefully and calmly near persons or farm animals. Also make sure that it is safe to reverse with the loader.

If the loader is not equipped with the boom self levelling:



WARNING

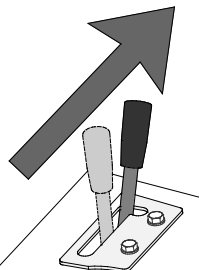
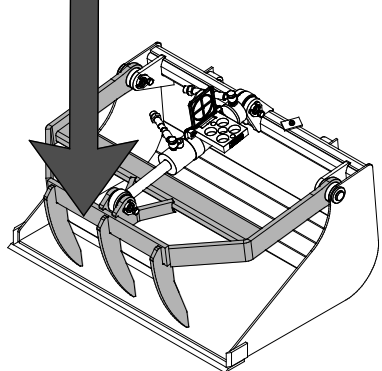
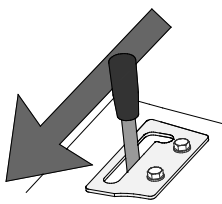
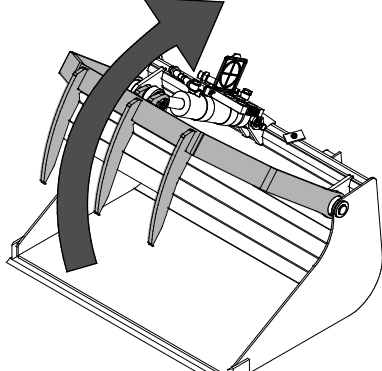
Prevent material from falling on you - Monitor the position of the bucket. When lifting material to a high level, the quick coupling plate must be simultaneously tilted to prevent the load from falling from on the loader or the operator. Fitting the loader with the boom self levelling option makes repeated loading work easier.



WARNING

Avoid sudden loss of stability - Never drag loads from e.g. a shelf. If the load that you want to lift from height is too heavy, the loader can tip forward when reversing with the loader. Never reverse and drag with the loader before making sure that the loader can handle the load that is being lifted.

Controlling the grapple:

	Move the auxiliary hydraulics control lever towards its locking position (or use the electric buttons of the optional electric joystick) to close the bucket. The lever should not be left to the locking position.	
	Move the lever away from the locking position to open the bucket.	

6.5 Material handling

Use maximum load indicated in the diagram of the operator's manual of the loader as a guideline. Remember that the actual load carrying capacity varies according to operating conditions, especially unevenness of operating area has a great effect. Be especially careful when the load sensor indicator is activated. Also, make sure that the ground surface is firm and follow recommended tire pressures.



WARNING

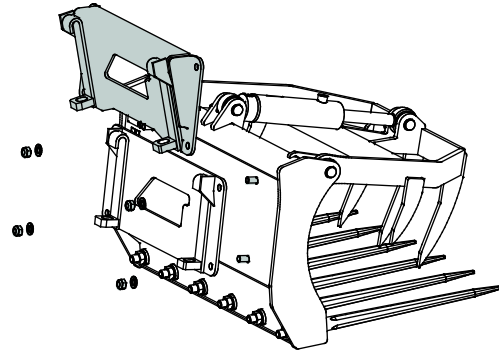
Risk of tipping over - Avoid overload and keep heavy loads close to the loader. Keep loads close to the ground when you are driving, especially during turns and uneven terrain. A heavy load or a long distance between the loader and the load's centre of gravity will affect the balance and handling of the loader. The use of additional counterweights is recommended especially when operating smaller loader models. Keep the telescopic boom retracted when moving the loader with a heavy load. Always wear a seat belt.

6.6 Options

6.6.1 Mounting plate with increased tipping angle A36337

The quick coupling plate is installed with bolts. With 500, 600, 700, and 800 series you can use the optional mounting plate with increased tipping angle when necessary. Use the mounting plate with increased tipping angle when you need a larger forward tilting angle with increased tipping angle when necessary. When the mounting plate with increased tipping angle is used, the tines of the silage forks can reach front tyres of the loader and damage them.

The mounting plate with increased tipping angle is installed on the silage forks in place of the original attachment coupling adapter.



NOTICE

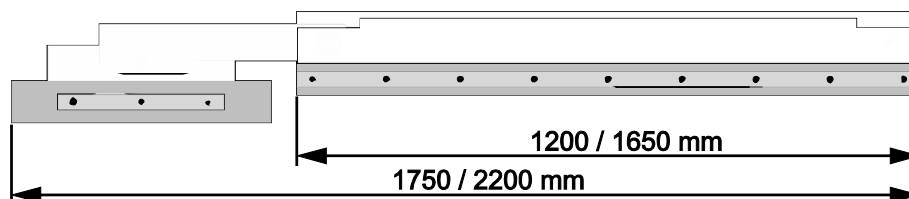
The mounting plate with increase tipping angle is not suitable for all loader models. Use it only in special cases where more forward tilt is necessary. When the mounting plate is installed, the tines can reach the front tyres and cause damage. Check the compatibility with your Avant dealer before use.



Risk of getting crushed under falling load - Make sure that the mounting plate is fastened securely. Check the tightness of the bolts. Tighten the bolts to 210 Nm. Do not use if any of the bolts are loose, damaged, or missing.

6.6.2 Feeding table cleaning blade A2936

With the help of the feeding table cleaning blade A2936, you can clear the cattle feeding table without demounting the silage forks from the loader. The cleaning blade is attached directly to the forks and it is locked by the grapple. It is not necessary to leave the driver's seat to attach the blade.



In narrow areas the side extension blade can be tilted onto the top of the main blade. The side extension is fitted on the right side of the blade, and it can be interchanged to the left side. As an option, a second extension blade is available for large areas.

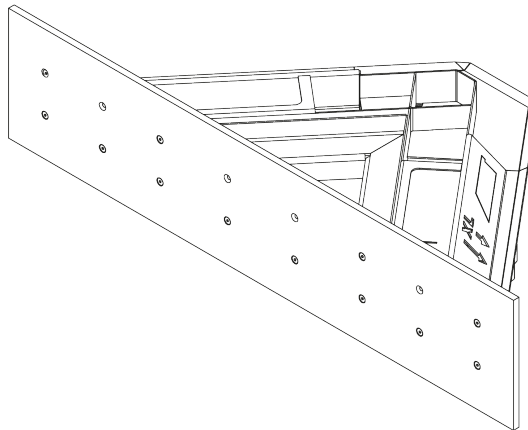
The cleaning blade is attached as follows:

1. Open the grapple and drive the tines towards the blade slot.
2. Keep the tines close to the ground and drive forward, until the tines have reached the end plate on the blade.
3. Close the grapple and ensure that the tines of the grapple fit the slot on the cleaning blade.



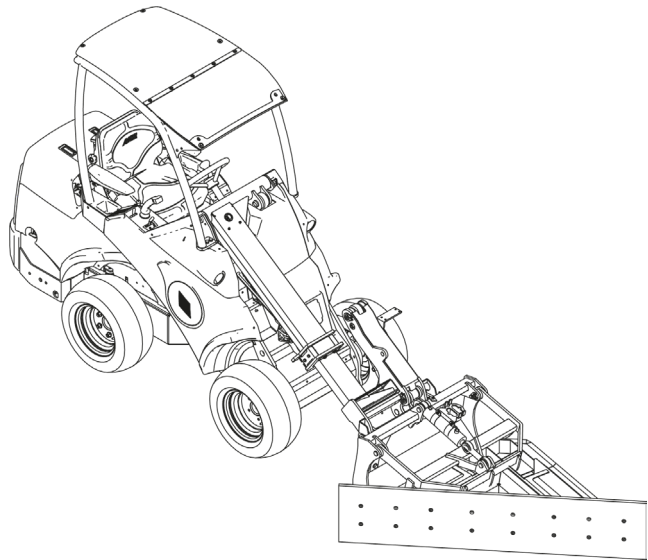
6.6.3 Silage dozer blade A37111

The silage fork is also possible to equip with a silage dozer blade. The blade is made of high-strength steel. In the bottom of the attachment, there is a durable nylon blade for equalizing the movement. The silage dozer blade A37111 is used for moving the uneaten silage back to the cattle on the feeding table.



Clearing the cattle feeding table can be done without demounting the silage forks from the loader. It is not necessary to leave the driver's seat to attach the attachment.

The silage dozer blade is mounted on the silage grab by closing the upper grapple on the dozer blade. The blade can be grabbed with the silage grab from two sides which means that it can move the silage both to the right and to the left.

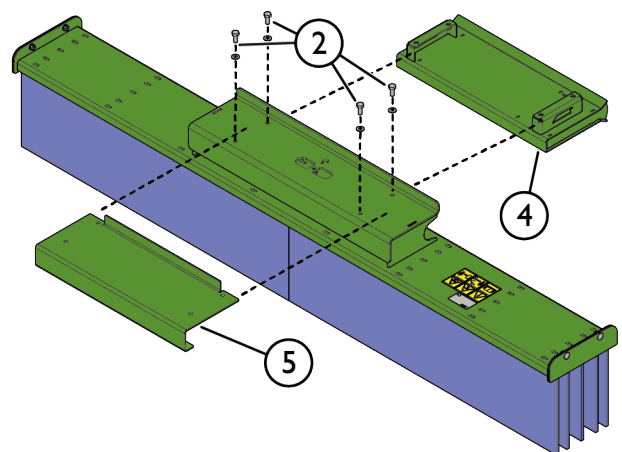
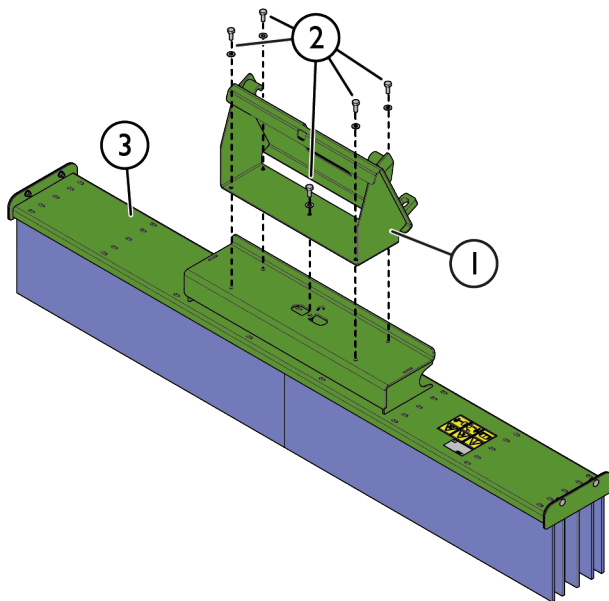


6.6.4 Push broom adapter A447080

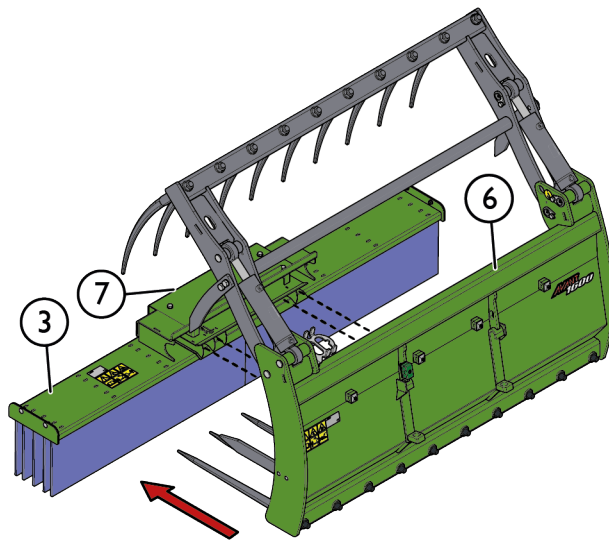
Starting to operate the push broom with Silage forks

Install the optional silage forks adapter by using existing fixing points.

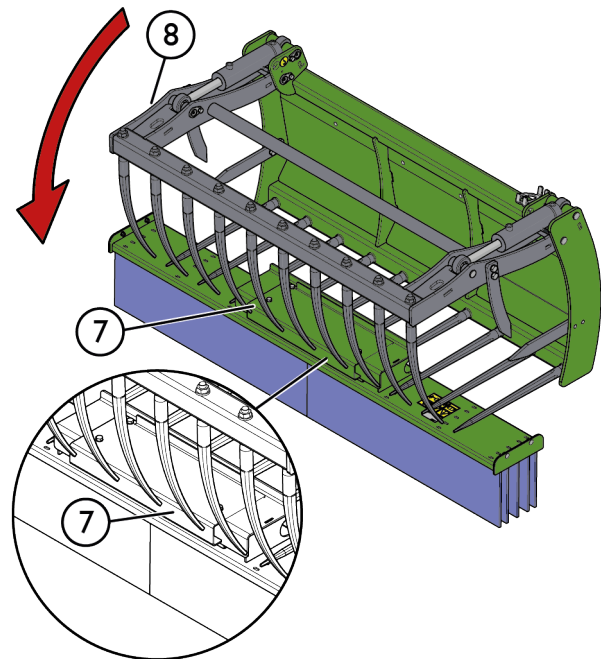
1. Remove the five bolts (2). Remove the standard coupling adapter (1) from the push broom (3).
2. Install the lower frame (4) of the silage forks coupling adapter into the push broom housing. Install the upper frame (5) onto the push broom adapter, as shown in the figure below. Attach the coupling adapter by using four of the bolts (2) removed in the previous step.



3. Drive the loader with the silage forks (6) close to the coupling adapter of the push broom (3). Insert the silage forks tines into the adapter (7) of the push broom (3), as shown in the picture below.



4. Lower the grapple (8) onto the quick coupling adapter (7) to lock the attachment, as shown in the picture below.



Using the push broom with silage forks

When the push broom is attached to the silage forks, operate it in the same way as when the broom is used alone. Tilt the attachment carefully. Small tilt movements can push the broom down too hard.

6.6.4.1 Compatibility of Avant push brooms with silage forks

All Avant push broom models are compatible with all silage fork models. However, it is recommended to use a push broom that is wider than the silage forks. The table below lists the available silage forks and push brooms and their widths.

Compatibility of silage forks and push brooms:

Silage forks	Push brooms		
	A449770, 1200 mm	A437121, 1500 mm	A447153, 2000 mm
A428328, 800 mm	•	•	•
A36299, 900 mm	•	•	•
A36313, 1050 mm	•	•	•
A36319, 1300 mm	(•)	•	•

6.7 Working on uneven ground

Extra caution is needed when using the equipment on inclined terrains and slopes. Drive slowly especially on inclined, uneven, or slippery surfaces, and avoid sudden changes in speed or direction. Operate the controls of the loader with careful and smooth movements. Watch out for ditches, holes on the ground, and other obstacles, as hitting an obstacle may cause the loader to tip over.

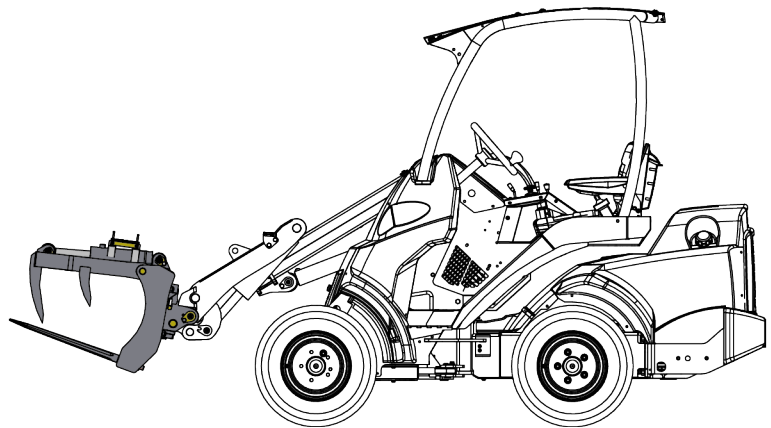
Maximum lifting capacity can not be achieved on inclined terrain. On horizontally tilted terrain the load must not be lifted high. The loader frame articulation should be kept straight when lifting heavy loads; turning the load during lifting operation will affect the stability of the loader and may lead to overturning of the machine.

6.7.1 Operating with a tilt adapter

When you work on slightly inclined terrains, a tilt adapter is an useful option. The tilt adapter is fitted in between the attachment and the quick coupling plate of the loader. The adapter allows the attachment to be tilted sideways. The possible tilt adapter options at the time of print of this manual are $\pm 12.5^\circ$ (model A34148), $\pm 45^\circ$ (model A36505) or $\pm 360^\circ$ (model A424406). Read the instructions provided with the tilt adapter and in the operator's manual of the loader. More information about the tilt adapters is available from your Avant retailer or from the Avant web pages at www.avanttechno.com.

6.8 Transport position

When driving with the loader, keep the attachment in the position shown in adjacent figure. Keep the grapple lowered completely and keep the silage forks as low and close to the ground as possible.



WARNING

Risk of tipping over - Keep the loader stable. Always transport the attachment as low and close to the ground as possible. Keep the telescopic boom retracted during transport drive.

6.9 Storing the attachment

Uncouple and store the attachment on level ground. The best way to store the attachment is a place where it is protected from direct sunlight, rain, and extreme temperatures.

- Avoid leaving the attachment laying directly on the ground. Place it on blocks of wood or on a pallet, for example.
- Place the multiconnector on to its holder as indicated with an arrow in the figure below. Never leave the hydraulic connector on the ground.
- Make sure that the attachment is secured against all movements during storage.
- Never climb on the attachment.

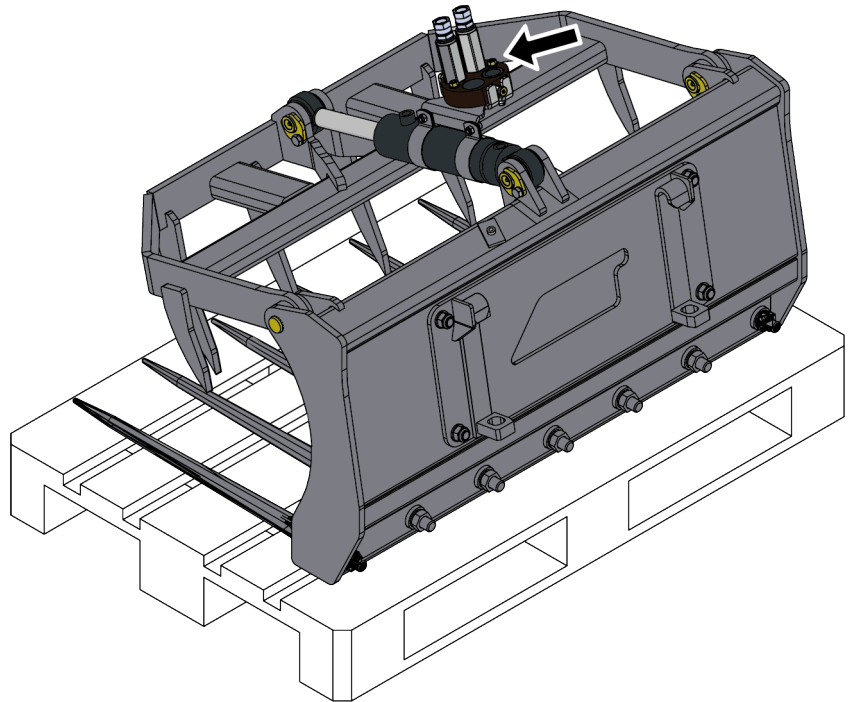
For longer storage periods touch up the paint where needed, grease the lubrication points and oil the visible parts of the hydraulic cylinder shaft to prevent rust damage.

Correct storage position:

To ensure the stability of the attachment during storage, always store it in position shown in adjacent figure.

Make sure that hydraulic hoses will not be left under the attachment.

Place the multiconnector to its holder on the attachment.

**NOTICE**

Avoid storing the attachment directly on the ground to prevent rust damage and for easier coupling procedure. Place the attachment on a pallet, for example.

7. Maintenance and Service

The attachment has been designed to be as maintenance free as possible. Continuous maintenance includes regular cleaning and lubrication, and monitoring the condition of the attachment. Because of the crushing hazard caused by lowering machine parts, all maintenance work must be done when moving parts have been lowered down completely and the attachment is lowered flat against the ground. Before any maintenance or service work follow the safe shutdown procedure.



DANGER

Risk of crushing - Never go under lifted attachment. Make sure the attachment is properly supported during all maintenance work. Never go under a lifted attachment. The loader boom may lower unexpectedly during maintenance causing serious injuries from crushing or impact even when the loader engine is not running. All maintenance and service must be performed when the attachment has been lowered down to a safe position.



WARNING

Risk of crushing - The grapple may lower by itself as a result of a fault or use or service mistake causing serious injuries. All maintenance procedures must be performed only when the grapple has been lowered down completely.



7.1 Inspection of hydraulic components

Check the condition of the hydraulic hoses and components when the engine has been turned off, system has cooled down, and the pressure has been relieved. Do not use the equipment, if you have discovered a leak in the hydraulic system of the attachment or the loader. Leaking hydraulic fluid may penetrate skin and cause serious injuries. Seek medical attention immediately in case hydraulic fluid penetrates the skin. Wash any part of body that has been in contact with hydraulic oil carefully with water and soap. Hydraulic fluid is also harmful to the environment and any leak to the environment must be prevented. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Operate the attachment only with type of hydraulic oil that is accepted for use in Avant loaders.



DANGER

Risk of high pressure fluid injection through skin - Release residual pressure before maintenance. Never handle hydraulic components when the hydraulic system is pressurised, since a fitting may break or become loose and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



Check hoses visually for cracks or abrasions. If there are signs of leaks, to check a component, hold up a piece of cardboard in the area where a leak is suspected. Do not use hands to search for leaks. Monitor the wearing of the hoses and stop the use, if the surface layer of any hose has worn off. Check the routing of the hoses; adjust the hose clamps to avoid abrasion to the hoses. The hoses have limited service life. Depending on operating conditions all hoses must be inspected thoroughly no later than after 3 to 5 years of use, and if required, they must be replaced with new ones.

Finding any fault means that the hydraulic hose or component must be replaced and the equipment must not be used until it is repaired. Spare parts are available from your nearest Avant retailer or authorised service point. Leave the repair work to professional service technicians, if you don't have adequate training, knowledge, and experience about hydraulic assemblies and how to perform the repairing safely.

7.2 Inspection of metal structures

Regularly inspect the metal structures of the attachment. Visually check for damage and inspect the quick coupling brackets and their surrounding area carefully. Do not use the attachment if it is deformed, cracked, torn or damaged in any other way. Regularly make sure that the the attachment and the bolts of the coupling plate are correctly secured

Welding repairs are only allowed to be carried out by professional welders. During welding repairs, only use methods and additives suited for the steel that is used in the attachment. For more information about repairs contact your nearest Avant service point.

7.3 Cleaning the attachment

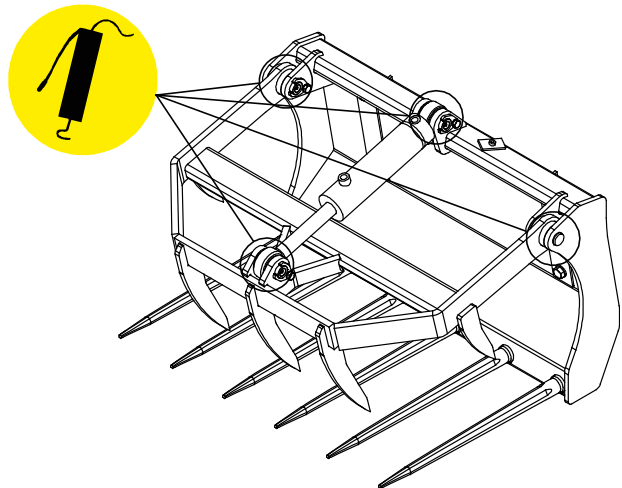
Clean the attachment regularly to prevent accumulation of dirt which is more difficult to remove. A pressure washer and mild detergent can be used for cleaning. Do not use strong solvents, and do not spray directly at the hydraulic components, or at the labels on the attachment.

7.4 Lubrication

There are a total of 4 lubrication points at the joints of the attachment. A small amount of grease should be added regularly.

Correct lubrication interval depends heavily on operating conditions but lubricant must be added at least after every 10 hours of use. Sufficient lubrication of the joints must be ensured, and if the joints have become dirty, lubricant must be added; added lubricant will push out dirt from joints.

Clean the end of the nipple before greasing and add only a small amount of grease at a time. All lubrication nipples are standard R1/8" nipples. Replace any damaged nipples.



7.5 End of life disposal

When the attachment is at the end of its useful life, recycle and dispose the attachment properly. Dismantle the attachment and separate the different materials, such as plastic, steel, rubber, and oils, for example. Drain and collect all oils and handle them by following the current local regulations. Never leave any oil or other material in the environment.

Recycle each material by delivering them to a proper collecting place. Let a waste management company do the recycling, if possible.

8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) from the date of purchase for the attachment it manufactures.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

FI EY-vaatimustenmukaisuusvakuutus
SV EG-försäkran om överensstämmelse
EN EC Declaration of Conformity

Alkuperäinen kieliversio
Originalversion
Original language



Valmistaja / Tillverkare / Manufacturer
Osoite / Adress / Address

AVANT TECNO OY
Ylötie 1
33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu /

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats /

We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:

SFS-EN ISO 12100, SFS-EN ISO 4413

Mallit / Modeller / Models

Avant	
Hydraulitoiminen rehupihti; Avant-kuormaajan työlaite	A428328
Hydraulisk Foder-/ensilagegrepp; arbetsredskap för Avant lastare	A36299
Hydraulic Silage forks; attachment for Avant loaders	A36313
	A36319



6.9.2024 Ylöjärvi, Finland

Jani Käkälä,
Toimitusjohtaja / Verkställande direktör /
CEO



AVANT[®]